2023-2027/CSE-DS/2/Group-5

FedXAI: Federated Multimodal Explainable Diagnosis

for Alzheimer's Disease Diagnosis

A Research Proposal Synopsis

(Based on Detailed Literature Review)

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

Session: 2023–2027

by

Pranjal Shukla (2300271640045)

Sahil Kumar (2300271540105)

Sneha Gupta (2300271540124)

Rishabh Dwivedi (2300271540096)

Under the Supervision of

Mr. Vivek Agarwal



AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

October 12, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

We also declare that the intellectual content of this synopsis is the product of our own work, even though we may have received assistance from others on style, presentation, and language expression.

We further declare that the research proposal outlined herein is based on an original review of existing literature conducted by us, and that any resulting publication (conference paper, journal article) or intellectual property (patent) arising from this work will duly acknowledge all sources and collaborators.

In accordance with the project rubric's emphasis on Academic Integrity, we commit to using plagiarism detection tools like Turnitin for self-checking throughout the writing process to ensure the originality of our work.

Name of Student	Roll Number	Signature
Pranjal Shukla	(2300271640045)	
Sahil Kumar	(2300271540105)	
Sneha Gupta	(2300271540124)	
Rishabh Dwivedi	(2300271540096)	

Session: 2023-2027 Date: October 12, 2026

CERTIFICATE

This is to certify that the work which is being presented in the B.Tech Research Proposal Synopsis entitled **"FedXAI: Federated Multimodal Explainable Diagnosis"**, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (2023-2027) and submitted to Ajay Kumar Garg Engineering College, Ghaziabad, affiliated with Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is an authentic record of our own work carried out under my supervision.

The matter presented in this synopsis, including the literature review, proposed methodology, and the planned research outcome (conference/journal publication or patent), has not been submitted by us for the award of any other degree of this or any other University.

Mr. Vivek Agarwal
Assistant Professor
Department of Computer Science and Engineering
Ajay Kumar Garg Engineering College, Ghaziabad

Abstract

The growing use of Artificial Intelligence in healthcare has created new opportunities for computer-assisted diagnosis. However, most models rely on a single modality, while clinical decisions typically draw on multiple sources of patient information. Centralizing such sensitive data for training also raises privacy and governance concerns, and deep learning models are often difficult to interpret, limiting clinical trust.

This research proposes FedXAI, a privacy-preserving, explainable, federated multimodal learning framework for **Alzheimer's Disease (AD) diagnosis** using paired structural brain MRI volumes and clinical data from the **Alzheimer's Disease Neuroimaging Initiative (ADNI)** dataset. The framework combines medical imaging with patient-specific clinical data using a **4-stage 3D Convolutional Neural Network** (3D CNN) as the image encoder and an MLP-based clinical encoder, fused to generate a unified disease prediction, without transferring raw patient data between institutions.

Federated Learning (via PyTorch and Flower) allows multiple simulated healthcare institutions to collaboratively train a global model using Federated Averaging (FedAvg), while retaining data locally. Each client's model update is clipped and perturbed with Gaussian noise once per round before transmission, as a practical privacy safeguard. For interpretability, **3D Grad-CAM** highlights important image regions and SHAP explains clinical feature contributions, together forming a multimodal explanation layer.

A key research focus is evaluating performance and explanation consistency under heterogeneous, non-IID data across institutions. The software is implemented and has run end-to-end on ADNI clinical data with synthetic placeholder MRI volumes. Expected outcomes include a trained multimodal model, a functional federated learning setup, multimodal explanations, experimental results, and an interactive Streamlit dashboard.

Keywords: Federated Learning, Explainable AI, Multimodal Deep Learning, 3D CNN, Alzheimer's Disease, ADNI, Medical Diagnosis, Grad-CAM, SHAP, Privacy-Preserving Machine Learning, Non-IID Data.

Contents

1 Introduction **7**
1.1 Background . . . 7
1.2 Problem Statement . . . 7
1.3 Aim . . . 8
1.4 Objectives . . . 8
1.5 Scope . . . 8

2 Literature Review **8**
2.1 Artificial Intelligence in Medical Diagnosis . . . 8
2.2 Multimodal Learning . . . 9
2.3 Federated Learning in Healthcare . . . 9
2.4 Explainable Artificial Intelligence . . . 9
2.5 Research Gap . . . 9

3 Societal Impact **9**
3.1 Healthcare Privacy . . . 10
3.2 Clinical Trust . . . 10
3.3 Collaborative Healthcare AI . . . 10
3.4 Responsible AI . . . 10

4 Universal Human Values **10**
4.1 Human-Centered Healthcare AI . . . 10
4.2 Privacy and Trust . . . 10
4.3 Transparency . . . 10
4.4 Responsibility and Limitations . . . 11

5 Methodology **11**
5.1 System Architecture . . . 11
5.2 Data Management . . . 11
5.3 Image Encoder . . . 11
5.4 Clinical Data Encoder . . . 11
5.5 Multimodal Fusion . . . 12
5.6 Federated Learning . . . 12
5.7 Explainable AI . . . 12
5.7.1 Grad-CAM . . . 13
5.7.2 SHAP . . . 13
5.8 Non-IID Data Evaluation . . . 13
5.9 Explanation Consistency . . . 13
5.10 Experimental Design . . . 13
5.10.1 Experiment 1: Centralized vs. Federated Learning . . . 13
5.10.2 Experiment 2: Effect of Data Distribution . . . 14

5.10.3 Experiment 3: Federated Optimization 14
5.10.4 Experiment 4: Explainability 14
5.11 Technology Stack 14

6 Expected Outcomes **14**
6.1 Multimodal Disease Prediction Model 14
6.2 Federated Learning Framework 14
6.3 Explainable Predictions 14
6.4 Performance Evaluation 15
6.5 Explanation Consistency Analysis 15
6.6 Interactive Dashboard 15
6.7 Research Contribution 15

7 Project Plan **15**
7.1 Phase-Wise Project Plan 15
7.1.1 Phase 1 — Literature Survey 15
7.1.2 Phase 2 — Dataset Selection 15
7.1.3 Phase 3 — Data Preprocessing 15
7.1.4 Phase 4 — Centralized Baseline 16
7.1.5 Phase 5 — Federated Implementation 16
7.1.6 Phase 6 — Explainability 16
7.1.7 Phase 7 — Experimental Evaluation 16
7.1.8 Phase 8 — Dashboard Development 16
7.1.9 Phase 9 — Analysis and Research Paper 16
7.2 Final Deliverables 16

1 Introduction

1.1 Background

Artificial Intelligence and Deep Learning have become important technologies in modern healthcare, particularly for computer-assisted disease diagnosis. Medical institutions generate large volumes of heterogeneous patient information, including MRI, CT and X-ray images, demographic information, clinical measurements, medical history, laboratory results, and diagnostic scores. These diverse sources of information can provide complementary evidence for disease prediction.

Traditional medical AI systems frequently focus on a single modality, such as medical-image classification. However, clinical decision-making is generally based on multiple sources of evidence. A multimodal learning approach can therefore combine image-derived representations with patient-specific clinical information to develop a more comprehensive predictive model.

Despite these opportunities, healthcare AI faces two major challenges. First, patient data is highly sensitive and cannot always be freely transferred between healthcare institutions. Centralizing data from multiple hospitals creates privacy, security, and governance concerns. Second, deep learning systems can operate as black boxes — a prediction such as a 94% probability of disease does not necessarily explain why the model produced that prediction.

FedXAI addresses these challenges by combining three major technologies: Multimodal Deep Learning, Federated Learning, and Explainable AI.

1.2 Problem Statement

Healthcare institutions possess valuable patient information that can be used to develop disease prediction systems. However, direct aggregation of such information into a centralized database may introduce privacy and security risks. Conventional machine learning approaches also struggle to exploit multiple modalities simultaneously while maintaining interpretability. Therefore, there is a need for a framework that can:

1. Combine medical imaging and clinical/tabular patient information.
2. Enable collaborative model training without transferring raw patient data.
3. Provide interpretable explanations for model predictions.
4. Handle heterogeneous data distributions across healthcare institutions.
5. Investigate whether explanations remain consistent across different participating institutions.

FedXAI proposes an integrated framework to address these requirements.

1.3 Aim

The primary aim of this research is to develop and evaluate a privacy-preserving, explainable federated multimodal learning framework for medical disease diagnosis.

1.4 Objectives

1. To develop a multimodal deep learning architecture combining medical images and clinical data.
2. To implement federated training using multiple simulated or real healthcare institutions.
3. To train a global model without transferring raw patient information to a central server.
4. To investigate Federated Averaging as the initial federated optimization strategy.
5. To apply Grad-CAM for explaining image-based predictions.
6. To apply SHAP for explaining clinical/tabular feature contributions.
7. To investigate the effect of non-IID data distributions among participating hospitals.
8. To evaluate the consistency and stability of explanations across federated clients.
9. To compare centralized and federated learning performance.
10. To develop an interactive Streamlit-based demonstration dashboard.

1.5 Scope

The disease application is **Alzheimer's Disease (AD) diagnosis** using paired structural MRI volumes and clinical data from the ADNI dataset. The research will focus on multimodal disease classification, federated model training, explainability, heterogeneous client distributions, and explanation consistency. The framework uses a **4-stage 3D Convolutional Neural Network** (3D CNN) for image representation, an MLP for clinical data, Flower for federated learning, and FedAvg as the baseline aggregation strategy.

2 Literature Review

2.1 Artificial Intelligence in Medical Diagnosis

Deep learning has become an important approach for medical image analysis because of its ability to automatically learn hierarchical representations from complex imaging data. CNN-based architectures can extract discriminative features from MRI, CT, and X-ray images and use these representations for disease classification. However, image-only systems may not utilise important clinical information available alongside medical images. This motivates the development of multimodal learning approaches. For Alzheimer's Disease specifically, Qiu et al. [11] demonstrated that combining structural MRI with clinical data significantly improves diagnostic accuracy over image-only models.

2.2 Multimodal Learning

Multimodal learning combines information from multiple data modalities to improve prediction. In the proposed system, medical images will be processed using a CNN-based encoder, while clinical information will be processed using a tabular neural network. Their resulting embeddings will be combined through feature fusion before classification. The proposed initial architecture uses intermediate feature fusion by concatenating image and clinical embeddings before passing them through fully connected layers.

2.3 Federated Learning in Healthcare

Federated Learning provides a distributed learning paradigm in which participating institutions retain their raw data locally while collaboratively training a shared model. The proposed system will simulate multiple hospitals, each independently performing local training and communicating model updates to a central server. FedAvg will initially be used to aggregate local model updates, and the global model will subsequently be redistributed to participating clients for further training rounds. Recent federated learning systems for Alzheimer's Disease include ADMarker [13], which detects digital AD biomarkers from sensor data using federated multimodal learning, and FedVC-ADDiM [14], which diagnoses AD from MRI scans distributed across different sites.

2.4 Explainable Artificial Intelligence

Explainability is particularly important in medical AI because clinicians require meaningful evidence supporting predictions. FedXAI will use modality-specific explanation techniques. For medical images, Grad-CAM will generate heatmaps identifying image regions that contribute to predictions. For clinical data, SHAP will estimate the contribution of individual features to a prediction.

2.5 Research Gap

Existing approaches may address medical image classification, multimodal learning, federated learning, or explainability independently. However, the proposed research focuses on their integration while additionally investigating cross-client explanation consistency. Healthcare institutions may have different patient populations, disease prevalence, and feature distributions; consequently, the data may be non-IID across clients. The proposed research will investigate whether such heterogeneity affects predictive performance, convergence, and explanation reliability — moving beyond predictive accuracy to examine whether explanations generated by a federated multimodal model remain stable and meaningful across different participating institutions. This gap is particularly pronounced for Alzheimer's Disease, where no prior work has simultaneously addressed federated training, multimodal fusion, and dual-modality explanation consistency under controlled non-IID conditions [12].

3 Societal Impact

3.1 Healthcare Privacy

Medical records and diagnostic images contain sensitive patient information. The proposed federated architecture keeps raw patient data within the institution where it was collected, allowing participating institutions to collaboratively train a model without creating a centralized repository of raw patient information. However, Federated Learning should not be considered an absolute privacy guarantee, since model updates can potentially reveal information; privacy-enhancing mechanisms may therefore be considered as future extensions.

3.2 Clinical Trust

A diagnostic AI system should provide more than a prediction score. Grad-CAM and SHAP can provide visual and feature-level explanations that help users understand the evidence contributing to a prediction.

3.3 Collaborative Healthcare AI

The framework can facilitate collaborative learning between institutions that may otherwise be unable or unwilling to pool their patient datasets, potentially allowing models to benefit from more diverse patient populations while maintaining local data ownership.

3.4 Responsible AI

The proposed system emphasises privacy, interpretability, evaluation across heterogeneous clients, and transparency — considerations that are important for the responsible development of AI-based healthcare technologies.

4 Universal Human Values

4.1 Human-Centered Healthcare AI

Healthcare technologies should ultimately support human well-being rather than merely optimise computational performance. FedXAI is designed around the principle that AI-assisted diagnosis should consider privacy, transparency, reliability, and responsible use.

4.2 Privacy and Trust

Maintaining patient data within healthcare institutions supports responsible handling of sensitive information. The federated architecture attempts to reduce the need for centralised raw-data sharing.

4.3 Transparency

Explainable AI contributes to transparency by providing information about the image regions and clinical features influencing predictions, supporting human understanding of AI-generated results.

4.4 Responsibility and Limitations

The system is intended as a research and decision-support framework rather than a replacement for qualified medical professionals. Model predictions and explanations must be evaluated carefully before any clinical deployment.

5 Methodology

5.1 System Architecture

The proposed FedXAI system will consist of participating healthcare clients and a central federated server. Each participating client will contain paired medical-image and clinical data, and the raw patient information will remain at the local institution.

5.2 Data Management

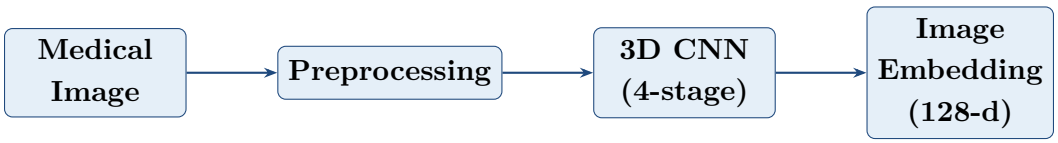
The data-management module will include:

- Medical-image preprocessing
- Clinical-data preprocessing
- Patient-level image/clinical pairing
- Training, validation, and testing splits
- Partitioning into simulated hospital/client datasets

The final disease and dataset will be selected based on the availability of an appropriate paired image and clinical dataset.

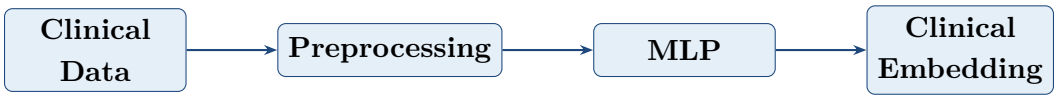
5.3 Image Encoder

Medical images will initially be processed using a **4-stage 3D Convolutional Neural Network (3D CNN)**, transforming an input volume into a 128-dimensional numerical feature representation.



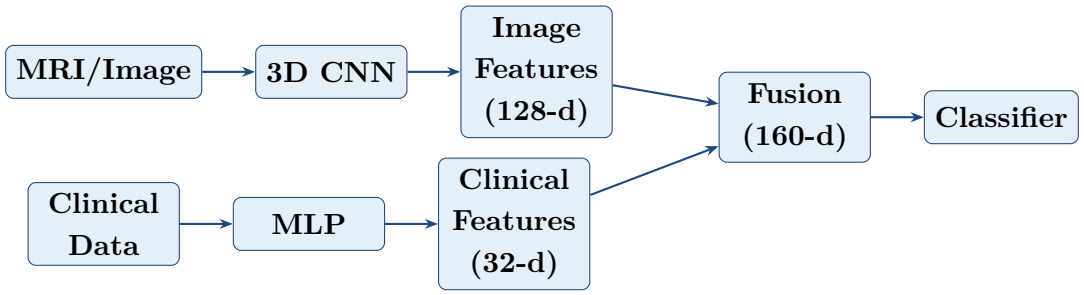
5.4 Clinical Data Encoder

Clinical information such as age, gender, medical history, and relevant measurements will undergo preprocessing and then be processed using a PyTorch-based MLP.



5.5 Multimodal Fusion

The image and clinical representations will be combined using feature-level fusion. For the initial implementation, a **128-dimensional** image embedding and a **32-dimensional** clinical embedding are concatenated into a **160-dimensional** representation.

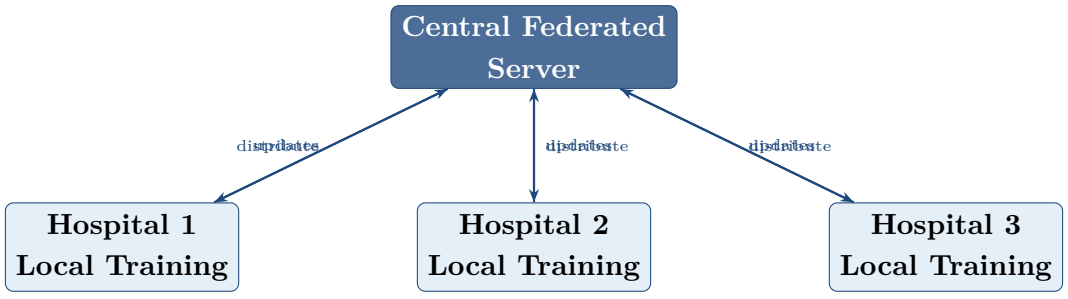


5.6 Federated Learning

Federated training will be implemented using Flower with PyTorch. The process will consist of:

1. Global model initialisation
2. Distribution of the global model to participating clients
3. Local training using client-specific data
4. Transmission of model parameters/updates — each client's update is **clipped to a maximum L2 norm (C_{dp}) and perturbed with Gaussian noise (σ) once per round** before transmission
5. Aggregation using FedAvg
6. Redistribution of the updated global model
7. Repetition for multiple federated rounds

The raw MRI images, clinical records, and patient identities will not be sent to the central federated server.



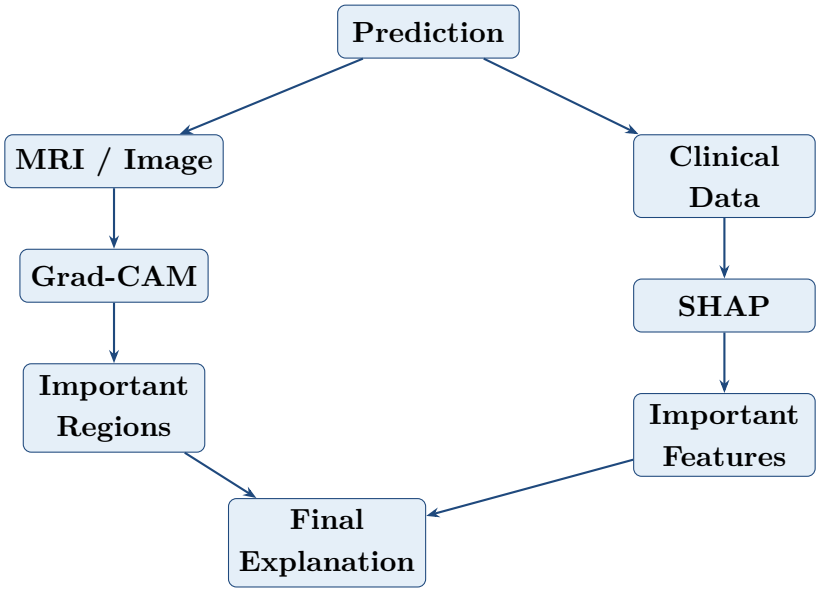
5.7 Explainable AI

5.7.1 Grad-CAM

Grad-CAM will be applied to the image branch to generate heatmaps highlighting image regions that contributed to the model prediction.

5.7.2 SHAP

SHAP will be applied to the clinical branch to estimate the contribution of individual clinical features.



5.8 Non-IID Data Evaluation

Different simulated hospitals will be assigned different data distributions, which may vary in patient demographics, disease prevalence, and other clinical characteristics. The study will evaluate global performance, individual client performance, model convergence, the effect of client heterogeneity, and explanation consistency.

5.9 Explanation Consistency

A major research component will be measuring whether explanations remain consistent across clients. For tabular data, feature rankings can be compared between hospitals. For image data, important regions identified by Grad-CAM can be compared where an appropriate reference or evaluation strategy is available. Potential measurements include feature-ranking similarity, explanation stability across federated rounds, local–global explanation similarity, and image-region overlap where an appropriate reference exists.

5.10 Experimental Design

5.10.1 Experiment 1: Centralized vs. Federated Learning

The centralized and federated models will be compared using Accuracy, Precision, Recall/Sensitivity, Specificity, F1-score, ROC-AUC, and Confusion Matrix.

5.10.2 Experiment 2: Effect of Data Distribution

Different simulated hospital distributions will be evaluated to determine their effect on local and global performance, convergence, and model variability.

5.10.3 Experiment 3: Federated Optimization

FedAvg will serve as the baseline. Depending on the final research scope, additional approaches such as FedProx or SCAFFOLD may be investigated.

5.10.4 Experiment 4: Explainability

The study will investigate important clinical features, important image regions, local versus global explanations, explanation stability, and cross-client consistency.

5.11 Technology Stack

Component	Technology
Programming Language	Python
Deep Learning	PyTorch
Federated Learning	Flower
Image Encoder	4-Stage 3D CNN (128-d embedding)
Tabular Encoder	PyTorch MLP
Image Processing	torchvision / OpenCV
Data Processing	Pandas, NumPy, Scikit-learn
Image Explainability	Grad-CAM (3D)
Tabular Explainability	SHAP (KernelSHAP)
FL Baseline	FedAvg
Dashboard	Streamlit
Version Control	Git + GitHub
Experiment Tracking	MLflow (Optional)
Deployment	Docker (Optional)

6 Expected Outcomes

6.1 Multimodal Disease Prediction Model

A trained model capable of combining medical imaging and clinical information for disease prediction.

6.2 Federated Learning Framework

A functioning federated learning environment in which multiple simulated healthcare institutions collaboratively train a global model without transferring raw patient data.

6.3 Explainable Predictions

The system will provide:

- Grad-CAM visualisations for medical images
- SHAP-based explanations for clinical features
- Combined multimodal interpretation of predictions

6.4 Performance Evaluation

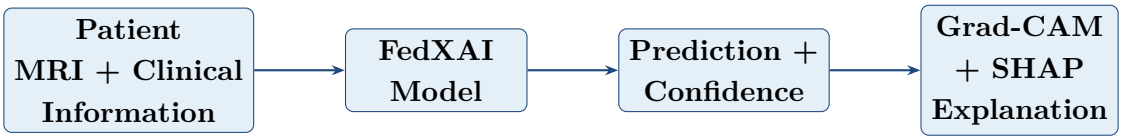
The project will provide quantitative comparisons between centralized and federated learning, as well as client-level performance analysis.

6.5 Explanation Consistency Analysis

The research will investigate whether the explanations produced by the global model remain stable and consistent across heterogeneous healthcare clients.

6.6 Interactive Dashboard

A Streamlit application will demonstrate the complete pipeline:



6.7 Research Contribution

The experimental results are expected to contribute to research concerning multimodal federated learning, heterogeneous healthcare data, privacy-preserving collaborative learning, and reliability of explanations across federated clients.

7 Project Plan

7.1 Phase-Wise Project Plan

7.1.1 Phase 1 — Literature Survey

Study existing research related to medical image classification, multimodal deep learning, federated learning, explainable AI, healthcare data privacy, non-IID federated learning, and explanation consistency.

7.1.2 Phase 2 — Dataset Selection

Identify an appropriate paired medical-image and clinical dataset. The final disease application will be selected based on dataset availability.

7.1.3 Phase 3 — Data Preprocessing

Develop pipelines for image preprocessing, clinical-data preprocessing, patient-level pairing, dataset splitting, and simulated hospital partitioning.

7.1.4 Phase 4 — Centralized Baseline

Develop and evaluate the initial centralised multimodal model.

7.1.5 Phase 5 — Federated Implementation

Implement the multimodal model using Flower and establish multiple simulated healthcare clients. Implement FedAvg as the initial federated baseline.

7.1.6 Phase 6 — Explainability

Integrate Grad-CAM for medical images and SHAP for clinical data.

7.1.7 Phase 7 — Experimental Evaluation

Conduct centralised vs. federated comparison, client heterogeneity experiments, federated optimisation experiments where applicable, and explanation stability and consistency analysis.

7.1.8 Phase 8 — Dashboard Development

Develop the Streamlit dashboard for demonstrating patient input, MRI upload, clinical information, prediction, confidence score, Grad-CAM visualisation, SHAP explanation, and federated training metrics.

7.1.9 Phase 9 — Analysis and Research Paper

Analyse the experimental results and document findings concerning multimodal learning, federated learning, heterogeneous healthcare data, and explanation reliability.

7.2 Final Deliverables

1. Multimodal disease prediction model
2. Federated learning implementation
3. Grad-CAM and SHAP explainability modules
4. Experimental evaluation results
5. Explanation-consistency analysis
6. Interactive Streamlit dashboard
7. Research paper based on the experimental findings

References

[1] H. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2017, pp. 1273–1282. https://arxiv.org/abs/1602.05629

[2] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, no. 2, pp. 336–359, 2020. https://doi.org/10.1007/s11263-019-01228-7

[3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774. https://arxiv.org/abs/1705.07874

[4] T. S. Chen, M. Dong, and B. Chen, "Multimodal deep learning for medical diagnosis: A review," *IEEE Transactions on Biomedical Engineering*, vol. 68, no. 8, pp. 2450–2462, 2021. https://doi.org/10.1109/TBME.2021.3053429

[5] D. J. Beutel et al., "Flower: A friendly federated learning research framework," *arXiv preprint arXiv:2007.14390*, 2020. https://arxiv.org/abs/2007.14390

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. https://arxiv.org/abs/1512.03385

[7] X. Li et al., "Multi-institutional deep learning modeling without sharing patient data: A feasibility study on brain tumor segmentation," *IEEE Transactions on Medical Imaging*, vol. 39, no. 9, pp. 2796–2806, 2020. https://doi.org/10.1109/TMI.2020.2992692

[8] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 2, 2020, pp. 429–450. https://arxiv.org/abs/1812.06127

[9] Office of the Controller General of Patents, Designs and Trade Marks, *Guidelines for Examination of Computer Related Inventions (CRIs)*, Department for Promotion of Industry and Internal Trade, Government of India, 2023. [Online]. Available: https://ipindia.gov.in/writereaddata/Portal/IPOGuidelinesManuals/1_86_1_Revised__Guidelines_for_Examination_of_Computer-related_Inventions_CRI__.pdf

[10] United Nations, *Transforming our world: The 2030 Agenda for Sustainable Development*, Resolution adopted by the General Assembly on 25 September 2015, A/RES/70/1. https://sdgs.un.org/2030agenda

[11] S. Qiu et al., "Multimodal deep learning for Alzheimer's disease dementia assessment," *Nature Communications*, vol. 13, Art. no. 3404, 2022. https://doi.org/10.1038/s41467-022-31037-5

[12] S. Jahan et al., "Federated explainable AI-based Alzheimer's disease prediction with multimodal data," *IEEE Access*, vol. 13, pp. 43435–43454, 2025. https://doi.org/10.1109/ACCESS.2025.3547343

[13] X. Ouyang et al., "ADMarker: A multi-modal federated learning system for monitoring digital biomarkers of Alzheimer's disease," in *Proc. 30th Annu. Int. Conf. Mobile Comput. Netw. (MobiCom)*, 2024, pp. 404–419. https://doi.org/10.1145/3636534.3649370

[14] I. Shahzad, J. Ouyang, and S. U. R. Khan, "FedVC-ADDiM: A federated learning framework for diagnosis of Alzheimer disease using deep learning," *Multimedia Systems*, vol. 32, Art. no. 161, 2026. https://doi.org/10.1007/s00530-026-02229-6